

# David Kurniadi Angdinata

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## Research interests

I am interested in the computational aspects of the Birch and Swinnerton-Dyer conjecture for abelian varieties over global fields and their formalisation in the Lean theorem prover. Please refer to my [website](#) for more details on my [research](#), [activities](#), [events](#), [talks](#), and [other projects](#).

## Academic record

- 10.25 – 8.27 **Senior Research Associate** in the University of East Anglia  
Grant: Scalable theorem proving via mathematical databases
- 9.21 – 9.25 **PhD Mathematics** in the London School of Geometry and Number Theory  
Thesis: L-functions of Dirichlet twists of elliptic curves: computations and congruences
- 10.20 – 6.21 **MASt Pure Mathematics** in the University of Cambridge
- 10.16 – 6.20 **MEng Pure Mathematics and Computational Logic** in Imperial College London
- 1.14 – 12.15 **Singapore-Cambridge GCE A-level** in Temasek Junior College
- 1.12 – 12.13 **Singapore-Cambridge GCE O-level** in Anderson Secondary School

## Employment record

- 3.26 – **Contractor** in AxiomMath
- 4.25 – 6.25 **Part-time Lean expert** in Harmonic
- 7.22 – 9.22 **Research assistant** in Huawei Technologies R&D UK Ltd
- 6.19 – 9.19 **Cryptography engineer** in Adjoint UK Ltd  
Developed and published the Haskell libraries `galois-field`, `elliptic-curve`, and `pairing`

## Research papers

- 5.26 ABC implies that Ramanujan’s tau function misses almost all primes  
In *Indagationes Mathematicae*
- 3.26 On the paucity of lattice triangles
- 1.26 Computing L-functions of  $\lambda$ -adic representations of global function fields
- 5.25 On L-values of elliptic curves twisted by cubic Dirichlet characters  
In *Canadian Journal of Mathematics*
- 10.24 Algebraicity of Artin–Hasse–Weil L-series over global function fields
- 7.23 An elementary formal proof of the group law on Weierstrass elliptic curves in any characteristic (with Junyan Xu)  
In *14th International Conference on Interactive Theorem Proving (ITP 2023)*, volume 268 of *Leibniz International Proceedings in Informatics*, pages 6:1–6:19. Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Wadern, Germany, 2023

## Mathlib contributions

- Elliptic curves The discriminant of a cubic equation, Weierstrass equations and their changes of variables, the group law in affine coordinates, the group law in Jacobian coordinates, the group law in projective coordinates, division polynomials and their degrees, the theory of elliptic divisibility sequences
- Ring theory maximal ideals and maximal spectra, the ring of S-integers of a Dedekind domain, the Selmer group of a Dedekind domain

\* 21 May 1997

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## Selected talks

- 15.7.26 Bridging Lean and the LMFDB  
*Thematic Program on AI and Mathematics: Formalization, Theorem Proving and Reasoning* in Seoul
- 26.3.26 Torsion subgroups of elliptic curves in Lean  
*SMS Spring Meeting: Formalization and Proof Assistants* in Brig
- 13.1.26 A blueprint for the Birch and Swinnerton-Dyer conjecture in Lean  
数学机械化重点实验室 *Seminar* in Beijing (Online)
- 4.9.25 Computing Dirichlet L-functions over global function fields  
Contributed talk for *Young Researchers in Algebraic Number Theory* in Nottingham
- 31.7.25 Rational points on elliptic curves in Lean  
Contributed talk for *Rational Points 2025* in Schney
- 25.6.25 Elliptic divisibility sequences in Lean  
Contributed talk for *British Mathematical Colloquium and British Applied Mathematics Colloquium* in Exeter
- 25.3.25 Teaching a computer algebraic number theory  
*Algebra, Number Theory, Logic and Representation Theory Seminar* in Norwich
- 3.3.25 Computing L-functions over global function fields  
Contributed talk for *Elliptic Curves in the Cotswolds* in Stonehouse
- 13.2.25 Algebraising foundations of elliptic curves  
*Formalisation of Mathematics with Interactive Theorem Provers* in Cambridge
- 17.1.25 Division polynomials of elliptic curves  
Contributed talk for *Lean Together 2025* in Zoom (Online)
- 5.9.24 Twisted elliptic L-values over global fields  
Contributed talk for *Algebraic Number Theory* in Munich
- 31.7.24 Denominators of BSD quotients  
Contributed talk for *Young Researchers in Algebraic Number Theory* in Oxford
- 26.6.24 Elliptic curves in mathlib  
*Formalising Algebraic Geometry* in Pasadena, CA (Online)
- 24.4.24 L-values of elliptic curves twisted by cubic characters  
*Linfot Number Theory Seminars* in Bristol
- 2.8.23 An elementary formal proof of the group law on Weierstrass elliptic curves in any characteristic  
*14th International Conference on Interactive Theorem Proving* in Białystok

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## Events organised

- 7–11.6.27 *Arithmetic of Curves, Galois representations, and L-functions: Celebrating the Mathematics of Tim and Vladimir Dokchitser*, event in Edinburgh  
Conference organised by Alex Bartel, Matthew Bisatt, Alexandros Constantinou, Júlia Martínez-Marín, Jordi Vilà-Casadevall, and myself
- 29.6–3.7.26 *Bridging Lean and the LMFDB*, event in Norwich  
Workshop organised by Chris Birkbeck and myself
- 22–25.7.24 *International Congress on Mathematical Software: Novel Formalisations of Mathematics in Lean*, event in Durham  
Conference session organised by Fangming Li, Amelia Livingston, Jujian Zhang, and myself

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## Selected conferences

- 15.3–9.4.27 *Building the Mathematical Library of the Future*, workshop in Berkeley, CA
- 25–29.1.27 *Bridging the LMFDB and Lean*, workshop in Cambridge, MA
- 13–24.7.26 *Thematic Program on AI and Mathematics*, workshop in Seoul
- 25–27.3.26 *SMS Spring Meeting: Formalization and Proof Assistants*, conference in Brig
- 27.7–2.8.25 *Rational Points 2025*, conference in Schney
- 7–11.7.25 *LMFDB, Computation, and Number Theory*, conference in Providence, RI (Online)
- 24–28.6.24 *Formalising Algebraic Geometry*, workshop in Pasadena (Online)
- 31.7–4.8.23 *14th International Conference on Interactive Theorem Proving*, conference in Białystok

\* 21 May 1997

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## Selected teaching

|             |                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Spring 2027 | Lectures for <i>Year 1 MTHA4008B Probability</i> in UEA                                                                                       |
| Spring 2026 | Lectures for <i>Year 1 MTHA4007B Computational Mathematics</i> in UEA                                                                         |
| Summer 2025 | Supervisions for <i>Year 1 Term 3 research projects</i> in UCL<br>Projects in Mordell's equation, Lean                                        |
| Spring 2025 | Problems classes for <i>Year 3 MATH60040 Formalising Mathematics</i> in Imperial                                                              |
| Summer 2024 | Supervisions for <i>Year 1 Term 3 research projects</i> in UCL<br>Projects in how to tell if a number is prime, group theory and Rubik's cube |
| Spring 2024 | Tutorials for <i>Year 3 6CCM351A Representation Theory of Finite Groups</i> in KCL                                                            |
| Spring 2024 | Lectures for <i>Year 2 MATH0034 Number Theory</i> in UCL                                                                                      |
| Summer 2023 | Supervisions for <i>Year 1 Term 3 research projects</i> in UCL<br>Projects in cryptography, sums of squares, the axiom of choice              |
| Spring 2023 | Tutorials for <i>Year 2 5CCM251A Discrete Mathematics</i> in KCL                                                                              |
| Autumn 2022 | Tutorials for <i>Year 2 5CCM224A Introduction to Number Theory</i> in KCL                                                                     |
| Summer 2022 | Supervisions for <i>Year 1 Term 3 research projects</i> in UCL<br>Projects in continued fractions, cryptography, Lean                         |
| Autumn 2021 | Tutorials for <i>Year 2 MATH0014 Further Linear Algebra</i> in UCL                                                                            |

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## Outreach activities

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|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Spring 2025 | <i>London Maths Outreach: Group Theory</i> in UCL<br>A 4-week introductory course for sixth form students led by Alberto Centelles                             |
| Spring 2025 | <i>London Maths Outreach: Rational Points and Elliptic Curves</i> in UCL<br>A 4-week introductory course for sixth form students led by Jed Thorpe             |
| Summer 2023 | <i>Logic and Proof Summer School</i> in UCL<br>A 5-day introductory course for sixth form students led by Nikoleta Kalaydzhieva                                |
| Spring 2023 | <i>London Maths Outreach: Elliptic Curves</i> in Harris Academy St John's Wood<br>A 4-week introductory course for sixth form students assisted by Ben Handley |

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## Awards attained

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|-------------|-----------------------------------------------------------------------------|
| 2025 – 2026 | LMS Early Career Fellowship (£10490, declined)                              |
| 2024        | MAPS Faculty Education Award 2024 for individual excellence (£500)          |
| 2021 – 2025 | Full funding for 4-year PhD research [EP/S021590/1]                         |
| 2020        | Governors' MSci JMC Prize for best overall performance in final year (£500) |
| 2020        | Donald Davies Prize for best final year individual project (£500)           |
| 2018        | Department of Mathematics UROP research studentship (£1200)                 |
| 2017        | G Research Ltd Prize for academic excellence (£500)                         |
| 2017 – 2020 | Faculty of Engineering Dean's List (2017, 2018, 2020)                       |
| 2012 – 2015 | MOE Full 4-year school-based scholarship                                    |

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## Language skills

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| Languages   | English, Mandarin/Hokkien, Indonesian/Malay, Japanese                 |
| Programming | Lean, Haskell, Python/SageMath, Magma, Java, C/C++, Prolog, PHP/MySQL |
| Tools       | LaTeX, XHTML/CSS, Git, Stack, Vim                                     |

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## Miscellaneous activities

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|-------------|---------------------------------------------------------------------------------------|
| 2022 – 2023 | Private tutor for mathematics and computer science in TutorChase and ElitePrep        |
| 2020 – 2021 | Owner and moderator of the Cambridge Part III Mathematics Discord server              |
| 2018 – 2021 | Live-TexEd lecture notes for geometry, algebra, and number theory available on GitHub |
| 2019 – 2020 | Problems curator for the Imperial College Mathematics Competition                     |
| 2018 – 2020 | Organiser for the Imperial College undergraduate mathematics colloquium               |

\* 21 May 1997